Noise Removal versus Signal Loss in Network Filtering of Daily GNSS Time Series

H. Oğuz Bolat

[a]*Department of Geomatics Engineering, Yıldız Technical University, Istanbul, Türkiye*

Abstract

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Keywords: GNSS time series, common-mode error, spatial filtering, transient deformation, signal preservation, deep learning

1. Introduction

Daily position solutions from the Global Navigation Satellite System (GNSS) reach formal uncertainties of about 1 mm in the horizontal components and 3–4 mm in the vertical. The Nevada Geodetic Laboratory (NGL) processed data from more than 17,000 GPS stations worldwide as of 2018 and distributes the resulting time series openly (Blewitt et al., 2018). Using these series to study crustal deformation, such as slow slip or postseismic afterslip, first requires removing noise, part of which is shared between stations.

Common-mode error is noise shared across many stations of a network. It arises from errors in satellite orbits and clocks, in the realisation of the reference frame and from large-scale environmental loading (Wdowinski et al., 1997; Dong et al., 2006; Kreemer and Blewitt, 2021). Several approaches have been used to estimate and remove it, from stacking (Wdowinski et al., 1997) and principal or independent component analysis (Dong et al., 2006; Gruszczynski et al., 2018; Zhou et al., 2022) to weighted averages and medians of neighbouring stations (Tian and Shen, 2016; Kreemer and Blewitt, 2021). These filters are usually evaluated by how much noise they remove, most often measured as the reduction in the root-mean-square (RMS) scatter of

Email address: oguz.bolat@std.yildiz.edu.tr (H. Oğuz Bolat)

the filtered series (Tian and Shen, 2016; Kreemer and Blewitt, 2021; Zhou et al., 2022).

Learned filters have been evaluated in different ways. Bachelot et al. (2025) measure noise reduction as the decrease in the average position of the whole network, which should be close to zero in the absence of common-mode noise, and assess signal preservation by comparing 280 slow-slip offsets, each the difference between the mean positions in the ten days after and before an event, measured on raw and denoised series. They note that without ground truth the denoising performance cannot be quantified in an absolute sense. Costantino et al. (2024) add synthetic slow-slip events to noise generated from real GNSS series by principal component analysis and Fourier phase randomisation, which preserves its spatial covariance and spectrum, and report the error of the recovered displacement as a function of the signal-to-noise ratio; on real data, they validate the result by its agreement with tectonic tremor. Mastella et al. (2025), whose method works on single stations, train on synthetic series and evaluate the noise removed from real series against the residual of a trajectory-model fit (GrAtSiD; Bedford and Bevis 2018), which they treat as the reference noise, while showing the preservation of a slow-slip transient by example.

The problem with judging filters by the reduction in RMS scatter is that it does not account for the amount of real transient signal removed. Kreemer and Blewitt (2021) note that the main reason their filter reduces scatter more than traditional stacking is that it absorbs local common-mode transients. How much small signal survives filtering affects the science. In Cascadia, Costantino et al. (2026) find that the moment–duration scaling, b-value and stress-drop distribution of slow-slip events all change with the detection threshold applied to slip inverted from denoised data, a threshold they describe as a proxy for the signal-to-noise ratio. The disagreement over a reported precursory acceleration of slip before large earthquakes (Bletery and Nocquet, 2023) turned on the same question, whether a common-mode correction removes only noise or also real deformation (Bletery and Nocquet, 2025).

Bachelot et al. (2025) is the closest prior work to this study. They predict each station from its eight nearest stations beyond 400 km, and chose the 400 km empirically, as the shortest distance at which slow-slip offsets in their study area were no longer reduced by denoising. They note that this distance is specific to their region and time window, and it is not clear how either the number of stations or the distance should change for other networks or for

transients of other sizes. Distance has been varied before: Tian and Shen (2016) estimated the common mode from stations beyond a series of increasing distance thresholds, and took the threshold at which the RMS changed most sharply as the approximate spatial extent of a regional transient. Their thresholds served to locate transients, however, and were judged by scatter, not by how much of a transient the filter removes.

In this study we measure signal loss directly, by planting transients of known spatial extent, amplitude and duration in the real residuals of 1,251 stations in and around California. Each filter predicts every station from N other stations, excluding all stations within a radius R of it, and the prediction is subtracted as noise. Both parameters are treated as experimental variables rather than constants: R varies from 0 to 400 km and N from 16 stations to the whole network. We run each filter twice, once on the original residuals and once with the transients added. The difference between the two cleaned series is what remains of the planted transient, and its ratio to the transient gives the fraction of signal the filter keeps. The same runs give the noise the filter removes, as the reduction in short-period scatter, and any spurious signal it creates at nearby stations where nothing was planted. Because the transients are planted in real rather than synthetic noise, these measurements have a ground truth without depending on how well a noise model reproduces real data. Testing geodetic methods with known synthetic signals is not new: Gualandi et al. (2016) evaluated source separation on simulated networks with known tectonic and volcanic sources in synthetic noise. What has not been measured is how the signal a common-mode filter removes depends on the size of the transient relative to the reach of the filter.

We apply this audit to filters from three families: target-inclusive filters, which include the target station in their own estimate (network stacking, sub-regional principal component analysis and a robust weighted median); distance-excluded filters, which exclude all stations within a fixed distance of the target, including the 400 km design of Bachelot et al. (2025); and leave-one-out filters ranging from weighted means and ridge regression to a learned kernel and a space–time attention network, the last also trained with planted transients. Realistic slip events on mapped faults and real earthquake offsets test whether the conclusions drawn from idealised transients hold for real signals. Section 2 describes the data and their processing, Section 3 the audit and its measures, and Section 4 the filters, which are listed in Table 1. Section 5 presents the results, and Section 6 discusses their implications for choosing and evaluating network filters.

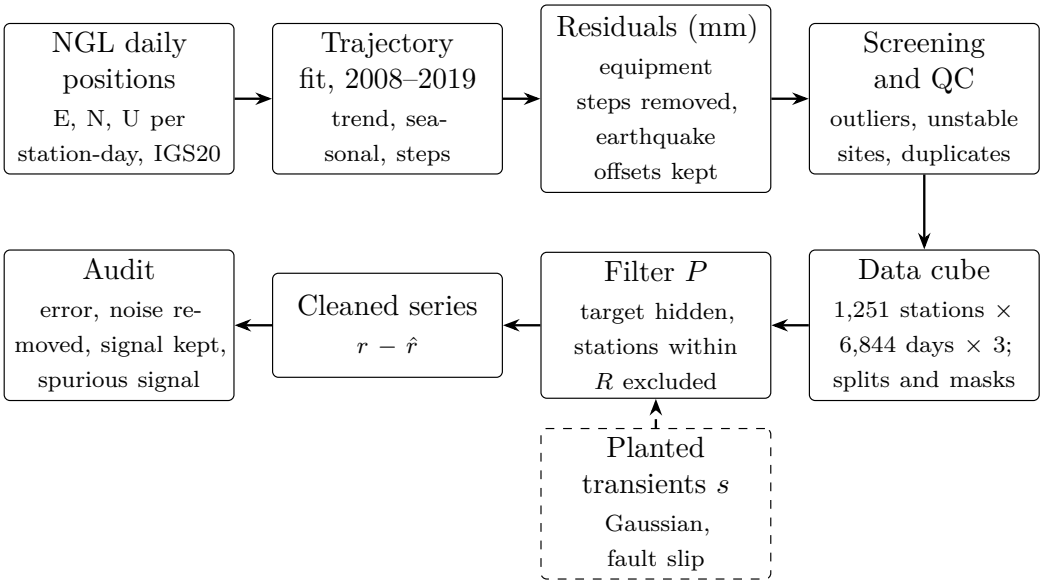


Figure 1: Processing chain from daily GNSS positions to the signal-loss audit.

2. Data and processing

Figure 1 summarises the processing chain. We take daily GNSS positions from NGL and reduce them to residual displacements by removing each station's long-term trajectory. The residuals are screened and arranged in a single array. Each filter then estimates the part of every station's residual that is shared with other stations, either from its N nearest neighbours, from all stations beyond a radius R, or, for target-inclusive filters, from stations including the target itself, and subtracts it. The cleaned series are scored on the noise removed, the planted transients and real offsets they keep, and the spurious signal they create. The rest of this section describes the data and their processing.

2.1. Data

The data are the daily IGS20 solutions of the Nevada Geodetic Laboratory (Nevada Geodetic Laboratory, 2026). For each station and day, a solution gives the position of the GNSS antenna in metres relative to a fixed reference point near the station, estimated from 24 hours of satellite observations, together with formal uncertainties of about 1 mm in the horizontal and 3–4 mm in the vertical. Each station therefore provides a three-component

(east, north and up) displacement series sampled once per day. In southern California the noise in these series is not only white but also temporally correlated, with flicker and random-walk components (Langbein, 2008), which is why we plant transients in the real series rather than in synthetic noise.

We use all stations between 32.4° and 42.1°N and between 124.6° and 114.0°W with at least 1,095 daily solutions: 1,381 stations in California and adjacent parts of Nevada, Oregon and Baja California. We restrict the analysis to 2008 onward, after the network had more than doubled in size from about 390 stations reporting on a typical day in 2005; since 2008, between about 820 and 1,030 stations report on a typical day. We remove 130 stations: 30 with less than one year of data after 2008; 22 with non-tectonic site motion such as groundwater-driven subsidence in the Central Valley, identified by a robust scale of the trajectory-model residual (Section 2.2) above 10 mm in the horizontal or 30 mm in the vertical; and 78 co-located duplicates within 1 km of a longer-running station. The remaining 1,251 stations (Fig. 2) provide 6.45×10^6 station-days. On 1 July 2018 the median distance to the nearest station was 9.7 km, and a typical station had 22 neighbours within 50 km and 73 within 100 km.

2.2. Residuals

Residuals are obtained by removing from each series a trajectory model of the station's long-term behaviour, fitted to that station's own data only; the residual on a given day is the difference between the observed position and this trajectory (Fig. 3a, b). The trajectory captures the predictable part of the motion, plate motion and seasonal cycles, so the residual retains transient deformation, whether local or regional, together with common-mode noise and the station's own noise. Plate motion is of the order of tens of millimetres per year, whereas transients are millimetres to centimetres in size; removing the known, predictable signals therefore makes the unknown transients visible.

The residuals are computed in four steps (Fig. 3). First, for each station i and component c, we fit by least squares the trajectory model

$$x_{i,c}(t) = a + v\,(t - t_0) + \sum_{k=1}^{2} \left[p_k \sin(2\pi k t) + q_k \cos(2\pi k t)\right] + \sum_j b_j\, H(t - t_j) + r_{i,c}(t), \tag{1}$$

where a is a constant, v a linear rate about the mean epoch t_0 of the fitted data, p_k and q_k the amplitudes of the annual ($k = 1$) and semi-annual ($k = 2$)

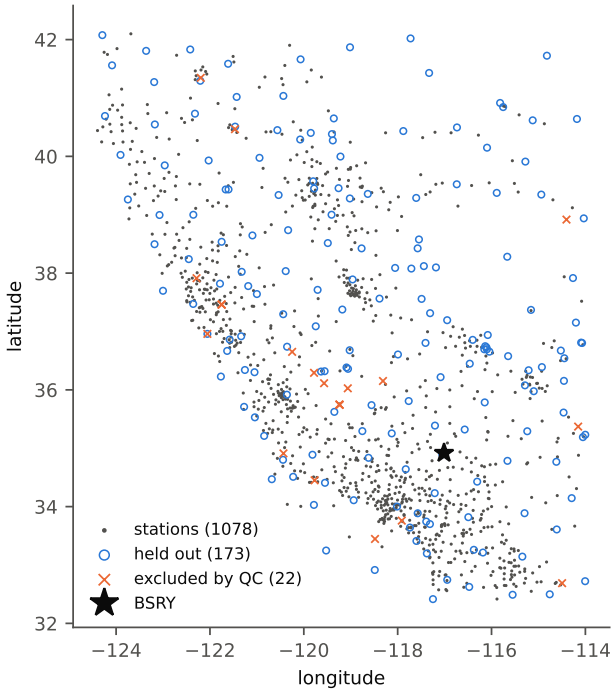


Figure 2: Stations used, held out and excluded.

harmonics, b_j the amplitudes of Heaviside steps H at the epochs t_j listed in the NGL step catalogue, and $r_{i,c}(t)$ the residual, with t in years. Only the training years (2008–2019) enter the fit. Second, the fitted trajectory is extrapolated to the later years, so that no validation or test day influences it. Third, steps caused by equipment changes are removed: those within the training years are estimated in the fit, and those after it with a local estimate of the jump, the median residual over the 30 days after the change minus that over the 30 days before. This local estimate uses days from the validation and test years, but no other station, and changes a series by at most a constant offset. Fourth, earthquake offsets are kept in the residuals, because they are spatially coherent deformation of the kind a filter should preserve; those within the training years are estimated in the fit, so that they do not bias the rate, and then added back.

Observations more than five robust standard deviations (1.4826 times the median absolute deviation) from a 61-day running median in any component are treated as missing, which removes 32,544 station-days, about 0.5% of

the data. Like the local equipment-step estimate, this screening uses days from all years, including the validation and test years, but no other station. Stations with unstable or non-tectonic motion are removed by the robust-scale criterion described in Section 2.1, applied to a trajectory fit from which earthquake offsets are also removed, so that real coseismic motion does not count against a station. Stations that develop site problems after the training years, identified by a more than threefold increase in day-to-day scatter or by a jump of more than 50 mm in the horizontal or 100 mm in the vertical with no catalogued step within 45 days, are kept as inputs but excluded from scoring; 19 stations are excluded this way. Because these criteria use only each station's own data and the step catalogue, the same stations are excluded for every filter.

The resulting residuals are arranged, together with their formal uncertainties and an availability mask, in an array of 1,251 stations by 6,844 days by three components. The years 2008–2019 are used for training and 2020–2021 for validation; the test years, from 2022 to 26 September 2026, are reserved for reporting, and days recorded after the method was frozen on 11 October 2026 will form a prospective test. A set of stations is held out from all training, and the interval from 1 July 2019 to 30 June 2020 at stations within 150 km of the 2019 Ridgecrest mainshock is excluded from training, so that no model is fitted or trained on its postseismic signal.

The held-out stations are used to test how well each trained filter generalises to stations it has not seen in training. They are chosen by farthest-point sampling (Gonzalez, 1985), which starts from a randomly chosen station and repeatedly adds the station farthest from all those already selected, so that the held-out stations are spread evenly across the network; 125 stations, 10% of the network, are selected this way. A further 56 stations are held out because their records in the training years are too short for the trajectory fit, with fewer than 365 daily solutions in 2008–2019; 8 stations fall in both groups, giving 173 in total.

We filter displacement rather than rates or daily increments, because differencing acts as a high-pass filter: it suppresses slow signals and amplifies short-period noise. For example, 2 cm of postseismic displacement accumulated over a year corresponds to a daily increment of about 0.05 mm, whereas the day-to-day scatter of the daily increments is about 1.4 mm in the horizontal components. The transients this study targets, slow slip and afterslip, evolve over days to months, so in daily increments they would lie far below the noise. Increments are used only as an additional input, alongside

displacement, to the attention network described in Section 4.4.

3. The signal-loss audit

3.1. Leave-one-out filtering and the exclusion radius

A filter P estimates the shared part of the residual of station i from N other stations, excluding all stations within R km of station i (Fig. 4); the cleaned series is $r - P(r)$.

Every filter that is scored on reconstruction error passes two leak checks, which compare predictions on inputs that differ only in data the filter is not allowed to use. In the first, the cells hidden for scoring are filled once with missing values and once with a poison value of 10^6 mm, while remaining marked as unavailable. In the second, the target station and every station within R of it are filled with the poison value, for five randomly chosen targets. In both checks the predictions must agree to within 0.001 mm. The checks are run for every filter at every radius, on the first year of the validation period, since any leak would also appear within a single year. Target-inclusive filters read the target station by design and are therefore not checked; for the same reason they are not scored on reconstruction error.

3.2. Measures of noise removed and of planted transients

Filters are compared on two measures of noise and two signal tests. The first noise measure, noise removed, is the reduction in short-period scatter at each station,

$$1 - \frac{\mathrm{rms}\Big(\mathrm{fast}(r - P(r))\Big)}{\mathrm{rms}\Big(\mathrm{fast}(r)\Big)}, \tag{2}$$

where the fast part of a series is the series minus its 61-day running median. It is averaged over the three components and summarised by its median over stations.

The second noise measure, reconstruction error, measures how closely the residuals of a station can be estimated from other stations. The error of each station, $r - P(r)$, is normalised by the station's robust noise scale, estimated from the training years only, and only its fast part (as in Eq. 2) is used. The reconstruction error is the root mean square of the normalised fast error, averaged over the three components, and summarised by its median over stations. A lower value means that the residuals of a station can be estimated more closely from other stations; predicting zero gives about 0.75 rather than

1, because separating the slow part also removes some of each station's own noise. Noise removed compares the scatter before and after filtering and can be computed for every filter. Target-inclusive filters, however, gain part of it by removing the station's own noise. They also cannot be scored on reconstruction error at all, since their estimate already contains the value being predicted (Section 3.1). Reconstruction error therefore compares the leave-one-out filters with each other, and it was also used to tune their hyperparameters on the validation years.

The first signal test, retention, is the fraction of a planted transient that remains after filtering. We add a transient s of known form to the real residuals and filter both the original and the modified residuals. The change in the cleaned series caused by the planted transient is

$$\Delta = \Big(r + s - P(r+s)\Big) - \Big(r - P(r)\Big), \qquad \rho = \frac{\langle \Delta,\, s \rangle}{\langle s,\, s \rangle}, \tag{3}$$

and the retention ρ is the fraction of the transient left in the cleaned series, computed over the horizontal components, the days of the transient and the stations within $2L$ of its centre, where L is its width (Section 3.3), and summarised by its median over transients. Plotting noise removed against retention for each filter and exclusion radius gives the trade-off curve between the two.

The second signal test, spurious signal, is how much of a planted transient leaks beyond its footprint. The planted signal of a Gaussian transient is zero beyond $3L$ from its centre, so the transient is confined to a footprint of radius $3L$. At stations outside the footprint whose cleaned series the transient can still change, nothing was planted, so any nonzero Δ is created by the filter. At each of these stations we take the peak horizontal amplitude of Δ as a fraction of the amplitude of the transient. We then keep the largest value among these stations, because that is where a spurious signal is most likely to be mistaken for a real one. As for retention, the median over transients is reported.

3.3. Planted and real signals

We plant Gaussian transients of known size,

$$s(d,t) = A \exp\!\left(-\frac{d^2}{2L^2}\right) g(t)\, \mathbf{u}, \tag{4}$$

where d is the distance of a station from the centre of the transient, A its amplitude, L its width, $\mathbf{u}$ a horizontal unit vector in a random direction and $g(t)$ its time profile. The profile rises over D days as a raised cosine, holds for 60 days and falls back to zero over another D days, so the transient leaves no permanent offset. Beyond $3L$ from the centre, s is set to zero. Every station in the footprint moves in the same direction, which is not realistic; these transients test the filters in an idealised case, in which the width L can be set exactly, and the fault-slip events below add realistic ones. The widths range from below to above the reach of a 16-neighbour filter at $R = 0$: the 16th-nearest neighbour of a station lies at a median distance of 40 km. The amplitudes A are 2, 5 and 10 mm, the widths L 10, 25, 50 and 100 km, and the durations D 10, 30 and 90 days. Each of the 36 combinations is planted at 20 centres, each at a randomly chosen scored station, giving 720 transients per filter. Their onsets fall in 300-day slots of the scored period, and the signal is added only on days with data. Many transients are planted in a single run of a filter. Two transients share a run only if no station's cleaned series can be changed by both, so the Δ at every station comes from a single transient. A larger exclusion radius lets a transient reach farther, so fewer transients fit in one run at larger radii.

A library of synthetic fault-slip events, shallow slow slip and afterslip, is also used. Synthetic events are computed from source models taken from published studies and moved to other faults of the same style in the SCEC Community Fault Model 7.0 (Plesch et al., 2007, 2024).

[Paragraph 8: real earthquake offsets and Ridgecrest.]

3.4. Protocol

All filter parameters were chosen on the validation split. The method was frozen on 11 October 2026 (repository tag freeze-2026-10-11), and all reported results are from the test years unless stated otherwise. The test years were not used for training or validation, but they were looked at on 7 October 2026, while the scoring rules were being revised, so they are not untouched; the days recorded after the freeze, which form the prospective test (Section 2.2), are. Four filters were run with five seeds each: the learned kernel over 16 neighbours, the learned kernel over 256 neighbours, the space–time attention network, and the space–time attention network trained with planted transients. The signal tests (Sections 3.2 and 3.3) used one seed of each.

4. Filters

Table 1 lists the filters. Leave-one-out filters never read the target's own values, on any day, whereas target-inclusive filters do. The "days used" column gives the days of the other stations' data that a prediction for day t reads.

4.1. Target-inclusive filters

Several target-inclusive filters were used. C1 is the stack of Wdowinski et al. (1997): the common mode for every station on a given day is the unweighted mean of all available stations on that day, the target included.

4.2. Distance-excluded filters

[Paragraph 3: the far stack and D1.]

4.3. Leave-one-out filters over nearest neighbours

[Paragraph 4: weighted means, robust medians and ridge regression.]

4.4. Learned filters

[Paragraph 5: training data and procedure.]
[Paragraph 6: the learned kernel.]
[Paragraph 7: the attention network and transient-augmented training.]

4.5. Implementation and use of AI tools

The pipeline described in this paper was implemented by Claude Opus 5.5 (Anthropic), used through Claude Code, between 6 and 12 October 2026. Conceptual and design choices were verified by the author, and all code and results were cross-checked. The tool was used to conceptualise the data-processing pipeline and to design and implement the common-mode and learned filters. All research questions are the author's own. The leave-one-out protocol was adopted from the literature (Bachelot et al., 2025), whereas the leak checks were designed specifically for this study by the tool under the author's direction (Section 3.1). All tests were verified by the author, [as were all code implementations ONLY ONCE DONE]. All cited works were read in full by the author. All analyses are reproducible: the code is openly available at `https://github.com/hogib/gnssdl` under the MIT licence. The method, including all design choices, hyper-parameters and scoring rules, was frozen on 11 October 2026 (repository tag `freeze-2026-10-11`), and all results reported for the test years (2022 to September 2026) were produced with the frozen method.

5. Results

5.1. The trade-off between noise removed and signal kept

[Paragraph 1: the overall trade-off (Fig. 6).]
[Paragraph 2: what the radius does.]

5.2. Number of stations versus distance

[Paragraph 3: count beats distance (Fig. 7).]
[Paragraph 3b: regression on the nearest stations (M3, M4).]

5.3. Realistic transients

[Paragraph 4: realistic transients, afterslip by distance (Fig. 8) and one sentence on shallow slow slip.]

5.4. Real earthquake offsets

[Paragraph 6: real offsets confirm the planted results (Fig. 9).]

5.5. Learned models

[Paragraph 7: the learned kernel.]
[Paragraph 8: more neighbours, more noise removed, less signal kept.]
[Paragraph 9: the attention network with and without planted-transient training.]
[Paragraph 9b: the attention network across exclusion radii.]
[Paragraph 9c: generalisation to held-out stations.]
[Paragraph 10 (optional): velocity repeatability.]

6. Discussion

[Paragraph 1: what practitioners should do.]
[Paragraph 2: why local filters remove the far field and why it matters.]
[Paragraph 3: relation to Bachelot et al.]
[Paragraph 4: what this means for deep learning.]
[Paragraph 5: limitations and future work.]

7. Conclusions

[Conclusions: write after the discussion.]

Data and code availability

[Data and code availability.]

Declaration of generative AI in the writing process

An AI assistant, Claude Opus 5.5 (Anthropic), was used to help design the benchmark, write the software and its tests, run the preliminary analyses and produce the figures, search and summarise the literature, and to draft, edit and improve readability of this text. The author directed the work, reviewed all AI-generated material and is responsible for the content.

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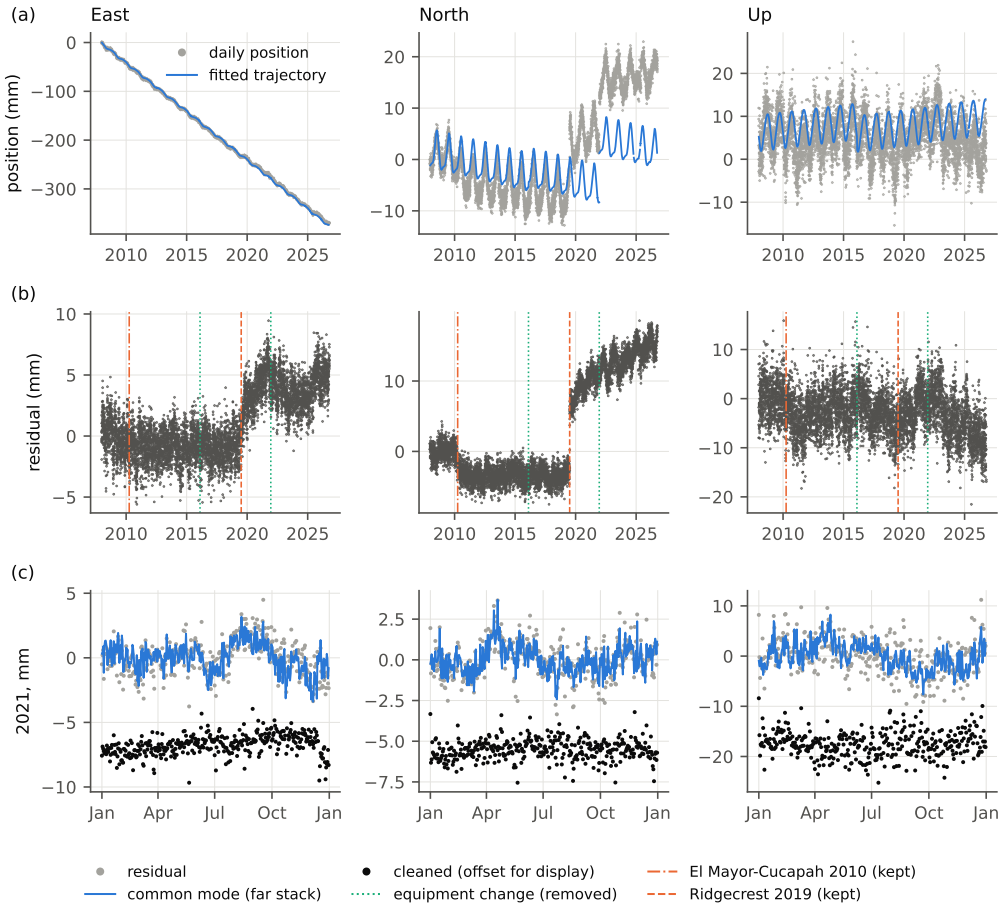


Figure 3: From positions to residuals at station BSRY. [Expand caption: (a) positions and trajectory, (b) residual, (c) one validation year with the network common mode and the cleaned series.]

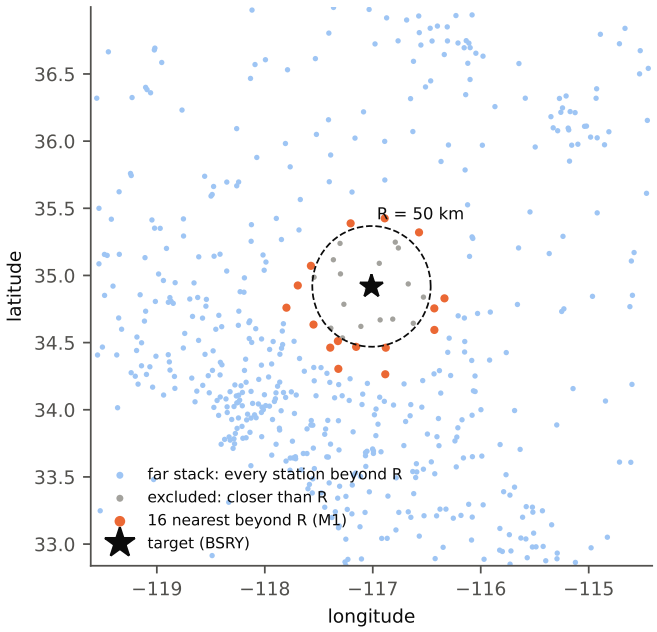


Figure 4: What a prediction for one target may use at $R = 50$ km.

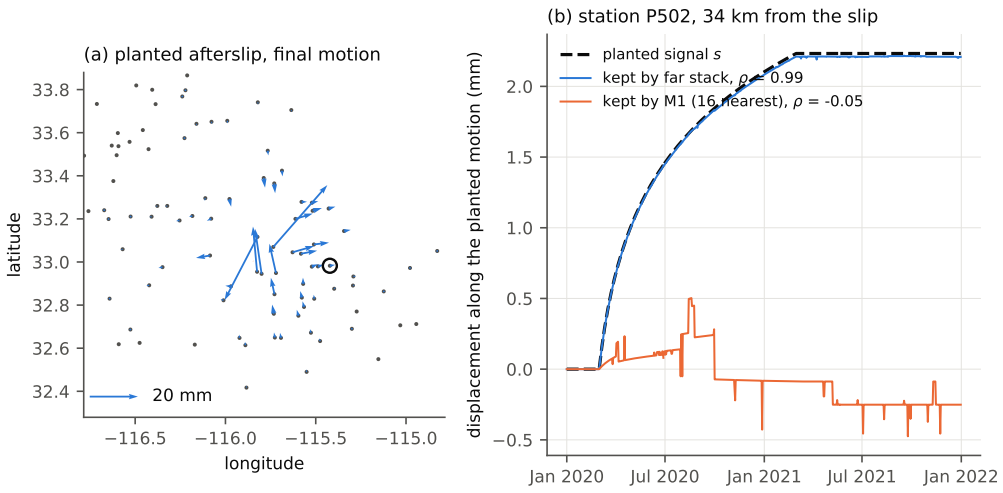


Figure 5: Measuring signal kept. [Expand caption; note that P502 is the median station among 17 at 15–40 km.]

Table 1: Filters audited. [Check descriptions.]

Name	Description	Parameters
Target-inclusive		
C1	Mean of the whole network, target included	0
C3	Sub-regional probabilistic PCA; 1, 4 or 8 regions × 1–3 components	∼3.7k/comp.
M2-self	Robust weighted median of the 16 nearest and the target	2
Distance-excluded		
Far stack	Mean of every station beyond R; D1 at $R =$ 400 km	0
Leave-one-out over nearest neighbours		
M1	Distance-weighted mean of the 16 (64, 256) nearest beyond R	1
M2	Robust weighted median of the 16 nearest beyond R	2
M3a / M3b	Ridge regression on 16 neighbours, per station / pooled	65/station; 257
M4a / M4b	M3 per time band (under 7, 7–60, 60–400, over 400 days)	4 × M3
Learned		
M3k	Learned kernel: neighbour weight from distance, azimuth and noise level; 16 or 256 neighbours	1,411
M5	Space–time attention: 32 neighbours and the far stack, 64-day window	212,048
M5-aug	M5 trained with planted transients in its inputs	212,048

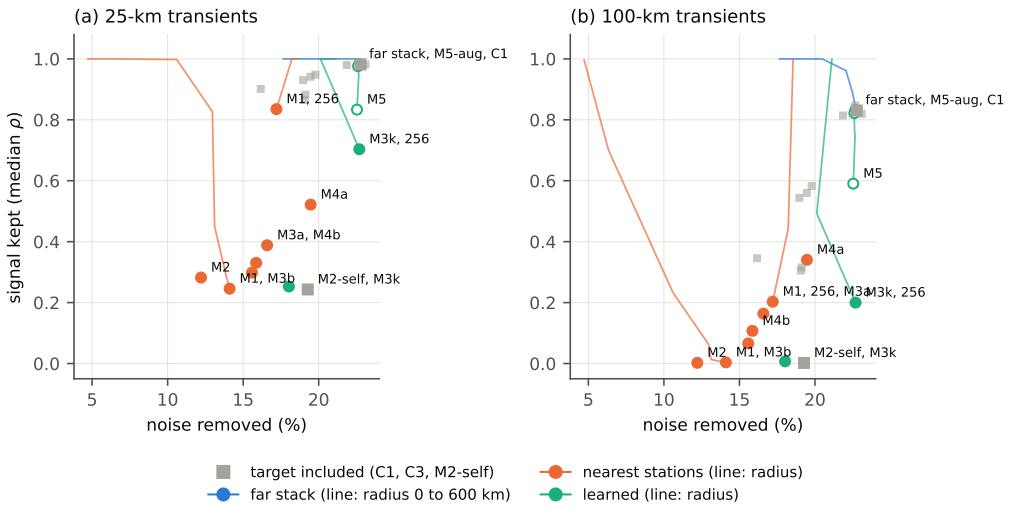


Figure 6: [Caption.]

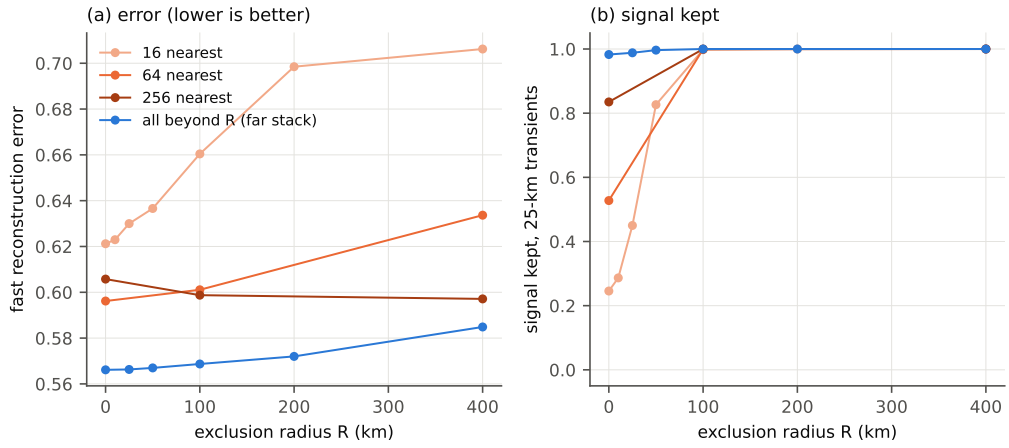


Figure 7: [Caption.]

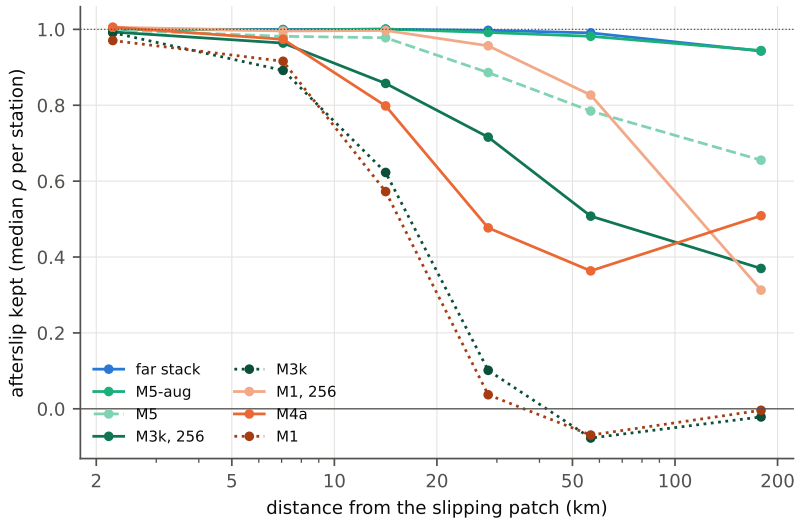


Figure 8: [Caption.]

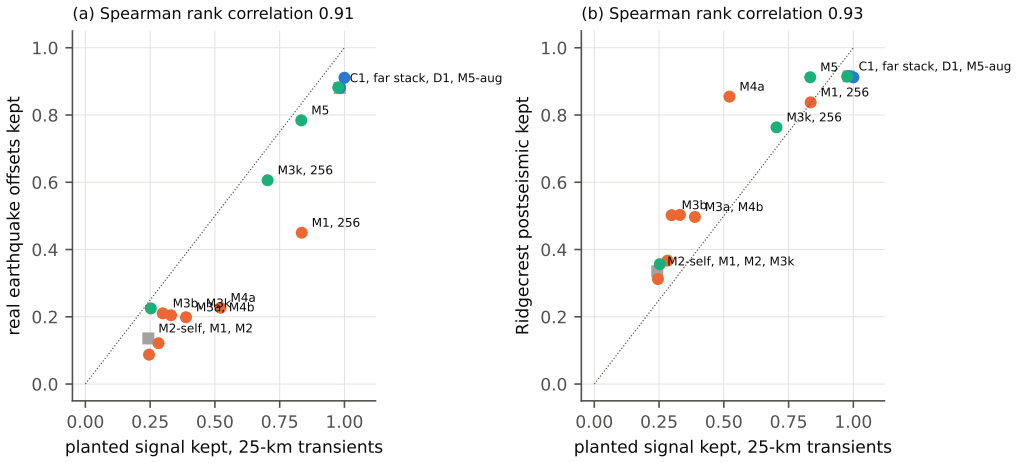


Figure 9: [Caption.]

Table 2: Main results at $R = 0$ unless stated, test years (2022 to September 2026). Error: fast leave-one-out error (lower is better; zero prediction 0.753). Kept: median retention of 25- and 100-km Gaussian transients; spurious: for 100-km transients; afterslip at 15–40 km from the patch; real earthquake offsets.

Filter	Error	Noise removed	Kept 25 / 100 km	Spurious	Afterslip	Real offsets
C1	–	22.8%	0.98 / 0.83	0.12	1.00	0.88
C3 (9 settings)	–	16.2–23.1%	0.86–0.99 / 0.31–0.85	0.24–1.58	0.97–1.00	0.67–0.87
M2-self	–	19.3%	0.24 / 0.00	0.11	0.18	0.14
Far stack	0.566	22.7%	0.98 / 0.83	0.12	1.00	0.88
D1 ($R = 400$ km)	0.585	20.5%	1.00 / 1.00	0.25	1.00	0.91
M1	0.621	14.1%	0.25 / 0.00	0.02	0.11	0.09
M1, 256	0.606	17.2%	0.84 / 0.20	0.11	0.96	0.45
M2	0.638	12.2%	0.28 / 0.00	0.12	0.20	0.12
M3a	0.603	16.6%	0.39 / 0.16	0.04	0.23	0.20
M3b	0.599	15.6%	0.30 / 0.07	0.04	0.24	0.21
M4a	0.582	19.5%	0.52 / 0.34	0.04	0.54	0.23
M4b	0.598	15.9%	0.33 / 0.11	0.04	0.33	0.20
M3k	0.595	18.0%	0.25 / 0.01	0.03	0.13	0.22
M3k, 256	0.562	22.7%	0.70 / 0.20	0.18	0.72	0.61
M5	0.563	22.5%	0.83 / 0.59	0.16	0.90	0.78
M5-aug	0.565	22.6%	0.98 / 0.82	0.16	0.99	0.88